BIG DATA ANALYTICS · COURSE PROJECT

Project Guide

Real data, Apache Spark, and recommendations a manager could act on

Item	Details
Marks	30 marks of the course total (separate from the labs)
Team	Pairs — 2 students
Timeline	Project pairs registration (28-10-2026), Project proposal (04-11-2026) and Project presentations (16-12-2026 or 13-01-2027)
Tools	PySpark and Parquet — in Google Colab or Databricks Free Edition

1. The project in short

In pairs, you take a **large real dataset**, process it with **Spark** (in Google Colab or Databricks Free Edition), analyse it, and end with **three recommendations** a manager could act on. You combine what the labs teach one at a time:

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Business question → data → Spark pipeline → analysis / prediction → insight → recommendation
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Finding a partner

Post on the course **Doubts Forum** to find a partner (say which project or topic interests you).
Register your pair and your dataset by the end of **28-10-2026**.
No partner by then? Contact the lecturer before the **29-10-2026** deadline.

2. Timeline and marks

Written work is due on the course platform by **Tuesday 23:59** of the date shown.

Date	Deliverable	What you submit	Marks
28-10-2026	Registration	Pair registration	—
04-11-2026	D1 · Proposal	Chosen project or own dataset (available online) PDF, max. 2 pages (checklist in Section 6)	5
First call 16-12-2026 or Second call 13-01-2027	D2 · Final report + code	PDF, max. 10 pages + notebooks + README	15
	D3 · Presentation	10 min talk + 5 min questions; both partners speak	10
	Total		**30**

3. What every project must have

Data

- **At least 1 million rows** (or 1 GB) of **real** data, legally usable — record the source, license and link.
- Do not submit the raw data — give the download link.

To do

#	To do	Module
1	**Spark for all big processing** (DataFrames or Spark SQL); pandas only for small results and charts	M3
2	**Raw → cleaned → curated** data stored as **Parquet**, partitioned by a sensible column	M4
3	**Data-quality log:** every problem found, how many rows, what you did and why	M6
4	**Analysis:** at least 3 KPIs, and one "why?" question investigated with evidence	M6
5	**One performance test:** e.g. CSV vs Parquet, caching, or partitions — with timings or explain ()	M3–M4
6	**Three recommendations** (who · what · why · how to measure) + limitations and ethics	M6
7	**Visualization:** at least 4 clear charts, each with a title, labelled axes and a one-sentence takeaway (e.g. KPI over time, comparison between groups, model result vs baseline)	M6

Plus one advanced option — choose one

Option	Hints
A · Machine learning	A Spark ML model that predicts something, compared with a simple baseline
B · Graph	Build a graph; PageRank or communities, and what they mean
C · Text	Sentiment, topics or classification of text
D · Streaming	Replay the data as a stream; windowed counts; compare with batch
E · Recommender	Spark recommender, compared with "most popular"; time-based split

4. Rules

- **No toy datasets** (Titanic, Iris, MNIST…) and **no lab datasets** from this course.
- **Not a pandas project:** processing the full data in pandas scores 0.
- **Cite** any code you reuse (Kaggle, GitHub, blogs).
- **Declare AI use** in one paragraph.
- **Both partners must** be able to explain every line.
- **Each pair must choose a completely different dataset (no sharing of datasets between pairs)**— first come, first served at registration.
- **Late work:** The presentation cannot be later than the **second call 13-01-2027**.

5. Choose a project — or bring your own data

Pick one of the 22 projects below or **search for another project (and use another dataset)** that interests you (see 5.2). Search for the dataset by its name (in bold) and check its size and license before you choose. Kaggle datasets need a free account.

5.1 Suggested projects

#	Project	About	Option
1	Flight delays	All US domestic flights with their delays and causes (**BTS On-Time Performance**). Analyse which airlines, airports, routes and hours have most delays. Predict whether a flight will arrive more than 15 minutes late, using only what is known before departure.	A
2	City service requests (311)	New York's 311 requests (noise, heating…) since 2010, with created and closed dates (**NYC 311 Service Requests**). Analyse complaint types and resolution times by agency and borough. Predict whether a request will take more than 7 days to close.	A
3	Hospital length of stay	All hospital stays in New York State, de-identified (**NY SPARCS Hospital Inpatient Discharges**): diagnosis, age group, admission type, hospital, length of stay. Analyse what drives long stays and predict how many days a new patient will stay — useful for bed planning. Add a short ethics note on health data.	A
4	Emergency response time	Every San Francisco fire and ambulance call, with the times it was received, dispatched and on scene (**SF Fire / EMS Dispatched Calls for Service**). Compute response times, find where and when they are slow, and predict whether a response will miss the city's target time.	A / D
5	Bike-share stations	Every trip of a city bike-share system (**Citi Bike** or **Divvy** trip data): start and end station, time, rider type. Find stations that run empty or full by hour, predict whether a rider is a member or a casual user, and suggest where bikes should be moved.	A or B
6	E-commerce funnel	An online shop's click log: product views, add-to-cart and purchases (**eCommerce behaviour data, REES46**, Kaggle). Find where shoppers drop out — by category, brand and hour. Predict whether a session will end in a purchase.	A
7	Customer value and baskets	Real sales from an online gift shop (**Online Retail II**, UCI) or a grocery app (**Instacart**, Kaggle). Find the best customers (how recently, how often, how much they buy) and which products are bought together. Predict how much a customer will spend next quarter.	A
8	Movie recommender	Millions of film ratings by users (**MovieLens 25M / 32M**). Study rating behaviour and the bias towards popular films. Build a recommender that suggests the top 10 films for each user, and compare it with recommending the most popular films.	E
9	Mortgage decisions and fairness	US mortgage applications, about 10 million a year (**HMDA loan-level data**): loan type, amount, income, region, decision. Analyse approval rates, predict approval, and check whether the model treats groups fairly.	A

#	Project	About	Option
10	Fraud detection *	Simulated mobile-money transactions with a fraud label (**PaySim**, Kaggle). Find fraud patterns by type, amount and time, train a classifier, and choose a decision threshold based on the cost of missed fraud vs blocked customers.	A
11	Product reviews	Amazon reviews for one or two product categories (**Amazon Reviews 2023**). Find which issues lead to low ratings and classify reviews as positive or negative from their text. Advise a product manager which problems to fix.	C
12	Local business reviews	Reviews, businesses and users in JSON (**Yelp Open Dataset**). Find what customers praise and complain about, by category and city, and predict a review's star rating from its text. Advise a restaurant owner what to fix first.	C (or B)
13	Music listening	19 million song plays by about 1,000 users, with timestamps (**Last.fm 1K**). Analyse listening habits by hour, day and artist, and recommend artists each user is likely to listen to next.	E
14	Hard-drive failure	Daily sensor (SMART) readings for hundreds of thousands of hard drives, with the day each one failed (**Backblaze Drive Stats**). Compare failure rates by model and age, and predict whether a drive will fail in the next 30 days. Failures are rare, and the train/test split must follow time.	A
15	Wikipedia navigation	How many readers went from article A to article B in a month (**Wikipedia Clickstream**). Build the click graph, find the central "gateway" pages (PageRank) and groups of related pages, and suggest missing links to editors.	B
16	Greener grocery basket *	Grocery orders (**Instacart**) plus average carbon footprint per food type (**Our World in Data**). Measure the footprint of baskets and aisles, and recommend lower footprint substitutes that shoppers are likely to accept.	E
17	Sustainable recipes *	Recipes and user ratings (**Food.com Recipes and Interactions**) plus food carbon data (**Our World in Data**). Estimate the CO_2 of each recipe, build a recipe recommender, and re-rank it to favour lower-carbon recipes — showing how much relevance is lost for the climate gain.	E + C
18	Sustainable fashion *	Purchases, articles and customers (**H&M Personalized Fashion Recommendations**, Kaggle). Tag product made from sustainable materials from their descriptions, and recommend the next product each customer will buy, giving more weight to sustainable ones.	E + C
19	Green AI *	Use MovieLens (or Goodreads) to study the recommender itself: train the same recommender model on different **amounts of data** (10%–100%) and with different **model sizes** (rank) and settings (iterations), measure accuracy against training time and estimated energy, and show where extra data or compute stops paying off.	E
20	Taxi demand (time series)	Every yellow-taxi trip in New York: zone, time, fare, tip (**NYC TLC Trip Records**, Parquet). Show when and where demand is highest and forecast the number of trips per zone in the next hour.	A / D

#	Project	About	Option
21	Public-safety patterns (time series)	Chicago's reported crimes since 2001: type, place, time, arrest (**Chicago Crimes 2001–present**). Analyse trends by area and time and forecast incidents per area next week. Requires a careful ethics discussion (bias, over-policing).	A
22	Smart-meter energy (time series)	Half-hourly electricity readings from thousands of London households (**Smart Meters in London**, Kaggle). Describe daily usage patterns by household type and forecast tomorrow's demand peak.	A

** Project 10 uses simulated (synthetic) transactions — accepted as an exception, because banks do not publish real fraud data. Projects 16–19 add a sustainability measure (e.g. kg CO_2e per recommendation) and show the trade-off between relevance and footprint.*

5.2 Your own dataset — welcome

You may use **any other dataset** — public open data, a Kaggle dataset, data from your workplace or another course — if:

- it meets the data rule in Section 3 (at least 1 million rows of real data);
- you are allowed to use it, and any personal data is anonymized;
- The lecturer approves it by the end of **04-11-2026** — send one paragraph: the source and link, the size, what you will analyze and what you will predict.

Good places to look: Kaggle, government open-data portals (city, national, EU), Google Dataset Search, Hugging Face datasets.

6. Checklists

Proposal (D1, max. 2 pages)

1. Pair names and IDs; project number or own dataset.
2. Business problem: who will use the results, for what decision?
3. What you will analyze, what you will predict, and the KPIs.
4. Data: source, link, license, size; where you will run it (Colab or Databricks).
5. Advanced option (A–E) and your plan; main risks.
6. Who leads which part (both partners code).

Final report (D2, max. 10 pages)

1. Summary: findings and recommendations.
2. Problem, questions, KPIs.
3. Data and pipeline: layers, Parquet, partitioning, data-quality log.
4. Analysis: KPIs and the "why?" question, with charts.
5. Advanced option and its evaluation against a baseline.
6. Performance test.
7. Three recommendations, limitations, ethics.
8. Appendix: AI-use declaration and who did what.

7. How you will be marked

Deliverable	Marks	Full marks when…
D1 · Proposal	5	Clear questions and KPIs; suitable data (size, license); realistic plan
D2 · Final report + code	15	Correct analysis; advanced option evaluated vs a baseline (3); performance test; recommendations; limitations, ethics, clear charts and code that runs
D3 · Presentation	10	Clear story from problem to action; honest evidence shown in clear charts; individual answers to questions

Both partners normally receive the same mark. The questions in D3 are answered **individually**, and the report appendix says who did what — a partner who did not contribute can receive a lower mark.